

Gabin An

- 🏢 Assistant Professor, Department of Computer Science and Engineering, Korea University
- 📍 Location: Seoul, Republic of Korea
- ✉ Email: gabin_an@korea.ac.kr
- 🎓 Google Scholar: <https://scholar.google.com/citations?user=T-58s1oAAAAJ>
- 🌐 Personal Website: <https://agb94.github.io/>
- 🌐 Lab Website: <https://ku-selene.github.io/>
- 🕒 Last updated on Aug 2, 2026

Employment History

- September 2025 – **Assistant Professor, Korea University**
 - Dept. of Computer Science and Engineering
 - Location: Seoul, Republic of Korea
- June 2024 – July 2025 **Software Engineer in Test, Roku**
 - Design, develop and maintain automation frameworks and tools
 - Location: Seoul, Republic of Korea

Education

- Mar 2020 – Aug 2024 **Ph.D., Computer Science, KAIST**, Republic of Korea
 - Advisor: Prof. Shin Yoo
 - Thesis: *Synergizing Fault Localization and Continuous Integration to Streamline Bug Resolution in Large-Scale Software Systems* (Distinguished Dissertation Award)
 - GPA: 4.11/4.3
- Mar 2018 – Feb 2020 **MSc, Computer Science, KAIST**, Republic of Korea
 - Advisor: Prof. Shin Yoo
 - Thesis: *Localising Software Faults by Learning Patterns of Failing Executions*
 - GPA: 4.15/4.3
- Feb 2012 – Feb 2018 **BSc, Computer Science, KAIST**, Republic of Korea
 - Major GPA: 4.06/4.3
 - Minor: Business and Technology Management
 - Honor: Summa Cum Laude
- Mar 2016 — Aug 2016 **Exchange Student, Informatik, TUM**, Germany

Selected Publications

* indicates equal contributions

- [1] J. Jung, C. You, J. Hong, T. Bach, and **G. An**, “Disassembly-Augmented Root Cause Analysis for C/C++ Segmentation Faults in SAP HANA,” in *41st IEEE/ACM International Conference on Automated Software Engineering (ASE’26)*, Oct. 2026, Industry Showcase, to appear.
- [2] T. Kiecker, E. Reichmann, H. Kang, **G. An**, and L. Grunske, “Is Three the Magic Number? An Empirical Evaluation of LLM-Based Repair Loops,” in *41st IEEE/ACM International Conference on Automated Software Engineering (ASE’26)*, Oct. 2026, NIER Track, to appear. arXiv: 2607.05197.
- [3] M. Kang, W. Oh, **G. An**, and H. Oh, “Pig: Leveraging Large Language Models for Python Library Migrations,” in *ACM International Conference on the Foundations of Software Engineering (FSE’26)*, Jul. 2026, Research Track, to appear.
- [4] S. Kang, S. Yun, J. Hong, S. Yoo, and **G. An**, “Finding the Needle in the Crash Stack: Industrial-Scale Crash Root Cause Localization with AutoCrashFL,” in *48th IEEE/ACM International Conference on Software Engineering (ICSE’26)*, Apr. 2026, SEIP Track, to appear.
- [5] H. Cho, S. Kang, **G. An**, and S. Yoo, “COSMosFL: Ensemble of Small Language Models for Fault Localisation,” in *2th International Workshop on Large Language Models for Code (LLM4Code@ICSE’25)*, May 2025.

- [6] N. Kim, S. Kang, **G. An**, and S. Yoo, "Lachesis: Predicting LLM Inference Accuracy using Structural Properties of Reasoning Paths," in *6th International Workshop on Deep Learning for Testing and Testing for Deep Learning (DeepTest@ICSE'25)*, May 2025.
- [7] H. Lee, **G. An**, and S. Yoo, "METAMON: Finding Inconsistencies between Program Documentation and Behavior using Metamorphic LLM Queries," in *2th International Workshop on Large Language Models for Code (LLM4Code@ICSE'25)*, May 2025.
- [8] J. Son, **G. An**, J. Hong, and S. Yoo, "Evaluating Machine Learning-Based Test Case Prioritization in the Real World: An Experiment with SAP HANA," in *18th IEEE International Conference on Software Testing, Verification and Validation (ICST'25)*, Mar. 2025.
- [9] **G. An**, J. Yoon, T. Bach, J. Hong, and S. Yoo, "Just-in-Time Flaky Test Detection via Abstracted Failure Symptom Matching," in *Proceedings of the 40th International Conference on Software Maintenance and Evolution (ICSME'24)*, Oct. 2024, Industry Track.
- [10] J. Choi, **G. An**, and S. Yoo, "Iterative Refactoring of Real-World Open-Source Programs with Large Language Models," in *16th International Symposium on Search-Based Software Engineering (SSBSE'24)*, Jul. 2024, Challenge Track.
- [11] S. Kang *, **G. An** *, and S. Yoo, "A Quantitative and Qualitative Evaluation of LLM-based Explainable Fault Localization," in *Proceedings of the ACM on Software Engineering and Volume 1, Number FSE (FSE'24)*, Jul. 2024, Research Track.
- [12] **G. An**, M. Kwon, K. Choi, J. Yi, and S. Yoo, "BUGSC++: A Highly Usable Real World Defect Benchmark for C/C++," in *Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23)*, Sep. 2023, Tool Demos.
- [13] J. Kim, **G. An**, R. Feldt, and S. Yoo, "Learning Test-Mutant Relationship for Accurate Fault Localisation," in *Information and Software Technology*, Jun. 2023.
- [14] **G. An**, J. Hong, N. Kim, and S. Yoo, "Fonte: Finding Bug Inducing Commits from Failures," in *Proceedings of the 45th IEEE/ACM International Conference on Software Engineering (ICSE'23)*, May 2023, Technical Track.
- [15] **G. An** and S. Yoo, "FDG: A Precise Measurement of Fault Diagnosability Gain of Test Cases," in *Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'22)*, Jul. 2022, Technical Track.
- [16] **G. An** *, J. Yoon *, J. Sohn, J. Hong, D. Hwang, and S. Yoo, "Automatically Identifying Shared Root Causes of Test Breakages in SAP HANA," in *Proceedings of the 44th IEEE/ACM International Conference on Software Engineering (ICSE'22)*, May 2022, SEIP Track.
- [17] **G. An**, J. Yoon, and S. Yoo, "Searching for Multi-Fault Programs in Defects4J," in *Proceedings of the 13th International Symposium on Search Based Software Engineering (SSBSE'21)*, Oct. 2021, Challenge Track.
- [18] J. Kim, **G. An**, R. Feldt, and S. Yoo, "Ahead of Time Mutation Based Fault Localisation using Statistical Inference," in *Proceedings of the 32nd International Symposium on Software Reliability Engineering (ISSRE'21)*, Oct. 2021, Research Track.
- [19] **G. An** and S. Yoo, "Reducing the Search Space of Bug Inducing Commits using Failure Coverage," in *Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'21)*, Aug. 2021, Ideas, Visions, and Reflections Track.
- [20] J. Sohn *, **G. An** *, J. Hong, D. Hwang, and S. Yoo, "Assisting Bug Report Assignment Using Automated Fault Localisation: An Industrial Case Study," in *Proceedings of the 14th IEEE International Conference on Software Testing, Verification and Validation (ICST'21)*, Apr. 2021, Industry Track.
- [21] **G. An**, A. Blot, J. Petke, and S. Yoo, "PyGGI 2.0: Language Independent Genetic Improvement Framework," in *Proceedings of the 2019 27th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'19)*, Aug. 2019, Tool Demos.
- [22] **G. An**, J. Kim, and S. Yoo, "Comparing Line and AST Granularity Level for Program Repair using PyGGI," in *Proceedings of the 4th Genetic Improvement Workshop (GI@ICSE'18)*, May 2018.

Awards and Achievements

- 2026  **ACM SIGSOFT Distinguished Paper Award; Distinguished Artifact Award**
 - ACM International Conference on the Foundations of Software Engineering (FSE 2026)
 - Paper: M. Kang, W. Oh, G. An, and H. Oh, “Pig: Leveraging Large Language Models for Python Library Migrations”
- 2025  **Distinguished Dissertation Award**
 - School of Computing, KAIST
- 2024  **Best Industry Paper Award**
 - The 40th International Conference on Software Maintenance and Evolution (ICSME 2024)
 - Paper: G. An, J. Yoon, T. Bach, J. Hong, and S. Yoo, “Just-in-Time Flaky Test Detection via Abstracted Failure Symptom Matching”
- 2022  **Best Paper Award**
 - Korea Software Congress (KSC 2022)
 - Paper: G. An, M. Kwon, K. Choi, and S. Yoo, “A Collection of Reproducible Bugs in C/C++ Programs” **PhD Fellowship Nomination**
 - Microsoft Research Asia
- 2019  **Best Short Paper Award**
 - Korea Conference on Software Engineering (KCSE 2019)
 - Paper: G. An, S. Yoo, “Search Space Reduction for Automated Program Repair Using Lexical Features”
- 2018  **2nd Place, Official Track**
 - CodRep’18: A Machine Learning Competition on Source Code Data
 - Organized by KTH Royal Institute of Technology, Stockholm, Sweden
 - <https://github.com/KTH/CodRep-competition>
- 2017  **Best Presentation Award**
 - Korea Software Congress (KSC 2017)
 - Paper: G. An, J. Kim, S. Lee, and S. Yoo, “PyGGI: Python General Framework for Genetic Improvement”

Miscellaneous Experience

Academic Services (PC & OC)

- 2027  **Proceedings Co-Chair**, the 24th International Mining Software Repositories Conference (MSR 2027)
 **Program Committee Member**, Research Track, the ACM International Conference on Foundations of Software Engineering (FSE 2027)
- 2026  **Program Committee Member**, Research Track/Journal-First Track, the 41th IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)
 **Program Committee Member**, Research Track, the 18th Symposium on Search Based Software Engineering (SSBSE 2026)
 **Industry Track Co-Chair**, 19th IEEE International Conference on Software Testing, Verification and Validation (ICST 2026)
 **Program Committee Member**, Industry Track, the ACM International Conference on Foundations of Software Engineering (FSE 2026)
- 2025  **Program Committee Member**, Tool Demonstration Track, the 32nd Asia-Pacific Software Engineering Conference (APSEC 2025)
 **Publicity Co-Chair**, the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025)
 **Program Committee Member**, Research Track/Journal-First Track, the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025)
 **Program Committee Member**, the 14th Genetic Improvement Workshop @ ICSE 2025
 **Program Committee Member**, Research Track, the 47th International Conference on Software Engineering (ICSE 2025)
- 2024  **Emergency Program Committee Member**, Research Track, the 39th IEEE/ACM International Conference on Automated Software Engineering (ASE 2024)
 **Program Co-Chair**, RENE/NIER Track, the 16th Symposium on Search Based Software Engineering (SSBSE 2024)

Miscellaneous Experience (continued)

- 2023
 - **Organising Committee Member**, the 13th Genetic Improvement Workshop @ ICSE 2024
 - **Program Committee Member**, RENE/NIER Track, the 15th Symposium on Search Based Software Engineering (SSBSE 2023)
 - **Program Committee Member**, Joint Artifact Evaluation Track and ROSE Festival, the 39th International Conference on Software Maintenance and Evolution (ICSME 2023)
- 2022
 - **Organising Committee Member**, the 12th Genetic Improvement Workshop @ ICSE 2023
 - **Web Chair**, the 14th Symposium on Search Based Software Engineering (SSBSE 2022)
 - **Program Committee Member**, the 11th Genetic Improvement Workshop @ GECCO 2022
 - **Program Committee Member**, Joint Artifact Evaluation Track and ROSE Festival, the 38th International Conference on Software Maintenance and Evolution (ICSME 2022)
- 2021
 - **Program Committee Member**, the 10th Genetic Improvement Workshop @ ICSE 2021
 - **Program Committee Member**, Artifact Evaluation Track, the 37th International Conference on Software Maintenance and Evolution (ICSME 2021)
- 2020
 - **Web Chair**, the 8th Genetic Improvement Workshop @ ICSE 2020

Academic Services (Journal Reviewing)

- ACM Transactions on Software Engineering and Methodology (TOSEM)
- IEEE Transactions on Software Engineering (TSE)
- Journal of Systems and Software (JSS)
- Automated Software Engineering (ASE Journal)

Teaching & Counselling Experience

- 2026
 - Lecturer, Software Verification (COSE419), Dept. of Computer Science and Engineering, Korea University, Spring 2026
- 2025
 - Lecturer, Algorithms (COSE214), Dept. of Computer Science and Engineering, Korea University, Fall 2025
- 2023
 - TA, Introduction to Software Engineering (CS350), School of Computing, KAIST, Spring 2023
- 2021
 - TA, Computer Ethics and Social Issues (CS489), School of Computing, KAIST, Fall 2021
 - TA, Operating Systems and Lab (CS330), School of Computing, KAIST, Spring 2021
- 2020
 - TA, Computer Ethics and Social Issues (CS489), School of Computing, KAIST, Fall 2020
 - TA, Automated Software Testing (CS453), School of Computing, KAIST, Spring 2020
- 2019
 - Head CA (Academic Counseling Assistant), School of Computing, KAIST, Spring 2019
- 2018
 - CA (Academic Counseling Assistant), School of Computing, KAIST, Fall 2018
 - TA, Programming Practice (CS109), KAIST, Spring 2018
- 2017
 - Undergraduate TA, Programming Practice (CS109), KAIST, Spring 2017

References

Available on Request